



On the large-scale radiative cooling induced by tropical cyclone activity

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Tropical cyclones (TCs) are among the most energetic phenomena in the climate system. Given their energetic nature, TCs induce upscale effects on the climate system, although these effects have not been extensively assessed. Understanding the influence of TCs on climate is important given the expected changes in TC activity with a warming climate. In this study, we provide evidence that TCs induce large-scale radiative cooling. We compare the climates of a coupled global climate model in two configurations: a control configuration with Earth-like TC activity and a perturbed configuration where TC activity is reduced by suppressing wind-induced surface heat exchange in areas with TC-like conditions. By comparing climate states, we find a reduction of incoming top-of-atmosphere radiation with more TC activity, largely in the subtropics. Globally, TC activity drives increases to longwave emission that outweigh increases to absorbed solar radiation. We show that decreases in sea surface temperature occur as an adjustment to TC-induced perturbations, coincident with increases in longwave emission caused by a drying of the free troposphere and a reduction in high cloud fraction. The results demonstrate the role of TCs in organizing convection, suggesting that TCs play a role in modulating climate sensitivity and large-scale energy transport.

tropical cyclones | Earth's energy budget | convective aggregation | deep convection | tropical climate

The climate system is regulated by the tropical energy budget. On average, the tropics receive a positive net annual radiative flux, which is balanced by poleward energy transport (1, 2). Given the importance of atmospheric moisture and tropical clouds in modulating energy in the climate system, it is critical to understand controls on their radiative properties. In the tropics, deep convection modulates the energy balance through condensational heat release, local and remote moisture changes, and the modification of cloud radiative heating. Additionally, the organization of deep convection regulates tropical climate remotely by drying the free troposphere, which enhances radiative cooling (3–8). There is also evidence that the degree of aggregation may change with a warming climate, although the direction and extent of this change are currently unclear (3, 9–11).

Tropical cyclones (TCs) are a ubiquitous type of organized deep convection. TCs generally originate in the tropics and subtropics, organize moisture and cloud cover into coherent multihundred kilometer structures, and can travel thousands of kilometers over several days. For these reasons, TCs are also powerful examples of convective aggregation (12–17). As with large-scale convective aggregation, it is anticipated that TC frequency and/or intensity will change with a warming climate. Despite a longstanding lack of consensus on how frequency will change, the peak intensity and rainfall of the strongest TCs are expected to increase (18–24).

Although TCs are small, short-lived, and infrequent relative to other organized deep convective phenomena, their energetic nature and ability to modulate cloud cover suggest they may have upscale impacts on the climate system (25–30). Despite a growing body of work investigating radiative properties of TCs and the effects of radiative feedbacks on TC evolution (31–33), the radiative effects of TCs on the climate have been less widely analyzed. This study focuses on this gap and asks the question: Does TC activity modulate energy into the climate system, and if so, through which mechanisms?

Several studies have provided insights into the direct and indirect radiative effects of TCs. The direct radiative effect of TCs has been shown to be negative, such that TCs reduce incoming top-of-atmosphere (TOA) radiation (25, 34). A framework developed to estimate localized TC radiative contributions using observational methods posited the potential for TCs to significantly modify the Earth's energy balance (34). This framework was then used to estimate TC-associated radiative cooling of 26 TW y^{-1} over

Significance

Although the effects of the large-scale climate on tropical cyclones (TCs) have been broadly explored, it is not yet fully understood how TC activity modifies the climate system. In this study, we analyze the upscale influence of TCs on the top-of-atmosphere energy budget to diagnose their effects on the large-scale environment using climate models. We show that TCs radiatively cool the climate system by organizing convection, cooling sea surface temperatures, drying the troposphere, and reducing high cloud fraction. The role of TCs in increasing large-scale convective organization is highlighted, which may shed light on the ability of TCs to modulate climate sensitivity and energy transport.

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a 20-y period due to the increased reflection of solar radiation due to TC high cloud tops (25). These findings align with direct radiative effects of TC-like vortices in idealized modeling studies, which show reductions in net TOA radiation due to anomalously high cloud tops despite a deeper moist column (7, 35–37). Idealized studies have also provided insights on the remote radiative effects of TC-like vortices, noting the positive correlation between vortex intensity and domain-averaged longwave cooling (13, 15, 35, 36, 38). The correlation is attributed to convective aggregation, as the consolidation of deep convection results in an increase of dry air domain fraction, thus promoting longwave cooling.

The indirect radiative effects of TCs are less clear. TCs reduce sea surface temperatures (SST) by generating cold wakes (39–41), which have been shown to reduce precipitation and cloud cover (42, 43). These reductions suggest that cold wakes locally modulate the radiative budget by decreasing cloud cover, which increases solar radiation at the surface and longwave emission to space at basin-scale (30). Despite mounting evidence that TCs induce radiative cooling across scales, the mechanisms connecting the local and remote radiative effects of TCs have not been explored globally over long timescales. Understanding this connection may be critical to determining how changes in TC activity will modulate the climate system.

In this study, we explore the extent of cross-scale TC-induced top-of-atmosphere (TOA) cooling using global climate models (GCMs), with an eye to the scale dependence of contributing mechanisms. First, we show that TC activity induces basin-scale radiative cooling in a coupled GCM. With more TCs, we find widespread SST cooling as an adjustment to radiative cooling from TC activity. The adjustment is moderated by widespread reductions in high cloud cover and a drying of the free troposphere, resulting in net cooling from increased longwave emission. We show that mechanisms inducing radiative cooling at basin-scale (e.g., longwave emission exceeding shortwave absorption) are opposite those inducing cooling local to TCs (e.g., shortwave reflection exceeding longwave trapping). Finally, we discuss the implications of these results for climate sensitivity and poleward energy transport in the context of TC activity changes.

Results

Reducing TC Activity in GCMs Using SWISHE. To investigate the upscale effects of TC activity on climate, we employ two configurations of each GCM used: a control configuration with Earth-like TC activity and a perturbed configuration with reduced TC activity.

The perturbed GCM configuration is generated by conditionally suppressing wind-induced surface heat exchange [termed *suppressing wind-induced surface heat exchange (SWISHE)*] at GCM grid cells with TC-like conditions (44) (850 hPa relative vorticity $> 1 \times 10^{-4} \text{ s}^{-1}$, lower and mid-tropospheric relative humidity (RH) $> 50\%$, and 10 m winds $> 15 \text{ m s}^{-1}$). SWISHE is implemented at each model timestep by reducing the wind speed used in the bulk flux parameterization to suppress evaporation. The difference for a field X between the control and SWISHE GCM configurations is denoted by the δ operator, where $\delta(X) = X_{\text{Control}} - X_{\text{SWISHE}}$, such that the sign of $\delta(X)$ corresponds to the influence of increased TC activity.

Two GCMs are used in three configurations within this study: a coupled atmosphere-ocean GCM (CM2.5-FLOR-FA, i.e. FLOR) and atmosphere-only GCMs (AGCMs, HiRAM and

AM2.5, the latter of which is the atmospheric component of CM2.5-FLOR-FA). We focus on findings from the coupled GCM, although results from AGCMs supplement this analysis. Given the nominal atmospheric model resolution of 0.5° , GCMs often use parameterizations for grid-scale cross-isentropic processes, such as those in TCs. Additional details on GCMs used are found in *Global Climate Models*.

In the SWISHE GCM configuration, TC frequency is reduced by 59% across GCMs used relative to control, with a multimodel reduction of $\sim 86\%$ for hurricane-strength TCs. The spatial distribution of TC activity and the differences in TC frequency between control and SWISHE configurations in FLOR are shown in Fig. 1. Details regarding the SWISHE methodology and simulated TC statistics are provided in *Modulation of TC Activity in Climate Models* and *SI Appendix, Fig. S2* (44).

Basin-Scale Radiative Cooling Driven by TC Activity. In a coupled atmosphere-ocean GCM (CM2.5-FLOR-FA), TOA net radiation is reduced where TC activity occurs (Fig. 2). Areas with TC activity (i.e., $5^\circ \times 5^\circ$ spatial bins with a minimum of 0.5 TC days per year, roughly 7% of global surface area) feature annual mean TOA radiative cooling ($\delta R_{\text{TOA}} = -0.40 \pm 0.05 \text{ W m}^{-2}$), while areas without TC activity feature a near-neutral area-averaged response ($\delta R_{\text{TOA}} = 0.03 \pm 0.01 \text{ W m}^{-2}$). The net radiative cooling associated with TCs ($\delta R_{\text{TOA}} = \text{absorbed shortwave radiation} - \text{outgoing longwave radiation} = \delta \text{ASR} - \delta \text{OLR}$) occurs from increases in longwave emissions exceeding enhanced absorption of solar radiation ($\delta \text{OLR} > \delta \text{ASR} \rightarrow \delta R_{\text{TOA}} < 0$). We find basin-scale radiative cooling occurs in an opposing way to the local radiative effect of TCs, where $\delta R_{\text{TOA}} < 0$ due to $\delta(-\text{ASR}) > \delta(-\text{OLR})$ (25).

Reductions in δR_{TOA} occur primarily over areas of TC activity, with maximum reductions reaching 3 W m^{-2} in areas with the strongest TC activity, suggesting TC-induced radiative cooling. However, areas outside of TC influence show neutral to modestly positive δR_{TOA} , which balances the radiative cooling in cyclonically active basins. Therefore, $\delta R_{\text{TOA}} > 0$ in areas outside of TC basins suggests that TC-induced radiative cooling is balanced by climate processes separate from TC impacts, pointing to the remote effects of TC activity.

To understand the radiative cooling associated with TCs, the contributing mechanisms can begin to be deduced from TC processes. Locally, TCs induce oceanic (i.e., SST cooling due to ocean mixing) and atmospheric perturbations (convergence in the boundary layer, divergence in the upper troposphere). In the environment immediately surrounding a TC, these processes suggest that radiative warming may be driven by weaker longwave emission due to surface cooling, while radiative cooling occurs due to a drier troposphere from low-level moisture aggregation and reductions in high cloud fraction from enhanced TC-induced subsidence, respectively. However, δR_{TOA} is also nonzero far from TC-active basins, indicating remote perturbations to the climate system from TC activity.

To diagnose potential contributors to radiative cooling, we decompose δR_{TOA} into its clear-sky ($\delta R_{\text{TOA, clear}}$) and cloud (i.e., cloud radiative effect, $\delta R_{\text{TOA, CRE}}$) components: $\delta R_{\text{TOA}} = \delta R_{\text{TOA, clear}} + \delta R_{\text{TOA, CRE}}$. In areas with TC activity exceeding 0.5 TC days per year, coupled GCM results show $\delta R_{\text{TOA, clear}} < 0$ and $\delta R_{\text{TOA, CRE}} < 0$, indicating that radiative cooling is attributable to both clear-sky and cloud contributions. Both quantities contribute similarly to δR_{TOA} in TC-active basins, as global mean $\delta R_{\text{TOA, clear}} = -0.19 \pm 0.04 \text{ W m}^{-2}$ and $\delta R_{\text{TOA, CRE}} = -0.21 \pm 0.03 \text{ W m}^{-2}$. To interpret these results,

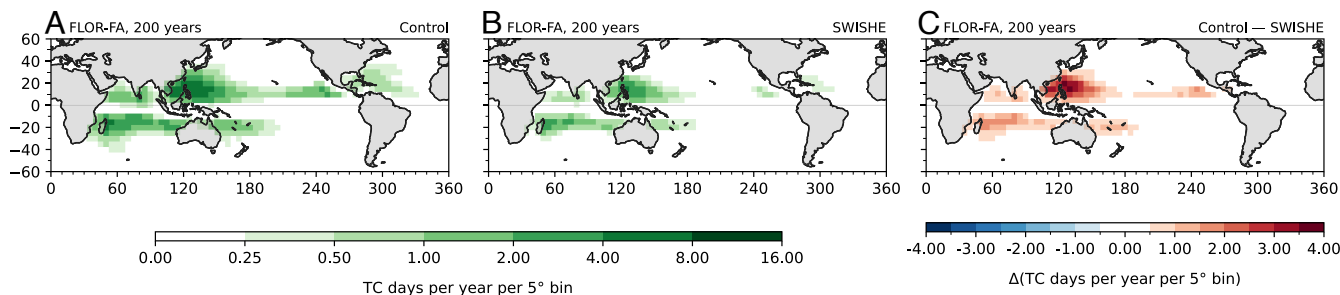


Fig. 1. Density maps of tropical cyclone (TC) locations for TCs of all intensities in CM2.5-FLOR-FA model integrations. Density is calculated per model grid cell by the number of days experiencing a TC per year, grouped by $5^\circ \times 5^\circ$ spatial bins. Each column corresponds to a different model configuration (control in panel A, SWISHE in panel B), and the difference (control—SWISHE) in panel C.

we investigate the physical mechanisms contributing to the clear-sky and cloud components of δR_{TOA} .

Widespread SST Cooling with More TCs. To understand why $\delta R_{\text{TOA, clear}} < 0$, we first analyze the role of TC-induced SST cooling. Coupled GCM runs show annual mean global $\delta \text{SST} = -0.07 \pm 0.01 \text{ K}$ (Fig. 3). In areas with TC activity exceeding 0.5 TC days per year, annual mean $\delta \text{SST} = -0.15 \pm 0.03 \text{ K}$, with localized reductions in δSST exceeding 0.5 K in regions with the highest TC activity. However, $\delta \text{SST} < 0$ in many areas outside of TC basins (Fig. 3), indicating that remote effects of increased TC activity induce SST cooling in these areas.

We analyze the extent of $\delta \text{SST} < 0$ by isolating the atmospheric contributions to δR_{TOA} and comparing of coupled and atmosphere-only GCM runs (SI Appendix, Fig. S3 A–C). AGCM runs show global mean $\delta R_{\text{TOA}} < 0$, similar to TC-active areas in the coupled GCM runs, as $\delta \text{OLR} > \delta \text{ASR}$. When the ocean is coupled to the atmosphere in CM2.5-FLOR-FA, the surface adjustment demonstrated by $\delta \text{SST} < 0$ contributes to energy balance restoration, as $\delta \text{SST} < 0 \Rightarrow \delta R_{\text{TOA}} > 0$, all else equal.

To begin approximating the contribution of δSST to $\delta R_{\text{TOA, clear}}$, we assume the ocean surface emits like a blackbody. Under this assumption, $\delta \text{SST} < 0$ indicates a decrease in surface emissions with TC activity, such that $\delta \text{SST} < 0 \Rightarrow \delta R_{\text{TOA, clear}} > 0$. Expectedly, global mean surface longwave emissions decrease with enhanced TC activity (i.e., decrease of

$0.43 \pm 0.02 \text{ W m}^{-2}$) and many areas featuring $\delta \text{SST} < 0$ outside of TC-active basins exhibit $\delta R_{\text{TOA}} > 0$. However, δSST and δR_{TOA} are not fully coupled inside nor outside of TC-active areas (compare Figs. 2 and 3), as many regions exhibiting $\delta \text{SST} < 0$ also feature $\delta R_{\text{TOA}} < 0$.

Insights into the decoupling of δSST and δR_{TOA} are provided by δR_{TOA} in AGCM runs, where $\delta \text{SST} = 0$. Radiative cooling in AGCMs (global mean $\delta R_{\text{TOA}} = -0.25 \pm 0.03 \text{ W m}^{-2}$ in AM2.5, which has an identical atmospheric model to CM2.5-FLOR-FA) is exceeded by warming from δSST in the coupled GCM runs. To offset the reduction in longwave emission by δSST , atmospheric processes modulate δR_{TOA} . Considering that $\delta \text{OLR}_{\text{clear}} > 0$ in TC-active basins for both AGCMs and the coupled GCM, it is evident that $\delta R_{\text{TOA, clear}}$ is not fully attributable to surface adjustment. As $\delta \text{OLR}_{\text{clear}} > 0$ in AGCMs, it is implied that a reduction in atmospheric moisture induces radiative cooling (45, 46). In summary, TC-induced SST cooling results in decreased surface longwave emissions, implying compensatory radiative cooling by atmospheric processes to maintain energy balance.

TCs Reduce Free Tropospheric Relative Humidity. Accordingly, TCs induce free-tropospheric drying in TC-active basins in all GCM runs. The global mean vertical profile of δRH in coupled GCM runs shows $\delta \text{RH} < 0$ in the free troposphere and slight $\delta \text{RH} > 0$ in the boundary layer (Fig. 4 A, Right subplot). A reduction in global mean free tropospheric RH

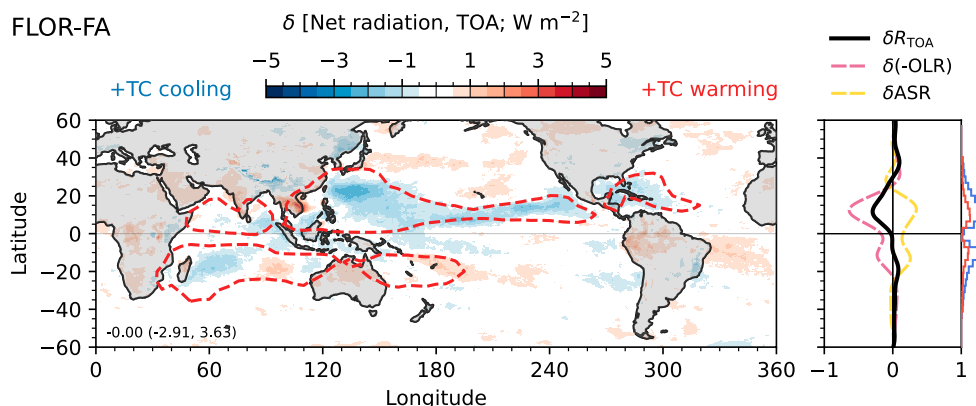


Fig. 2. Annual-mean TOA net radiation difference between control and SWISHE experiments (δR_{TOA}) for CM2.5-FLOR-FA. Time-mean differences are shown in the Left column and zonal-time mean differences of δR_{TOA} and its shortwave (absorbed shortwave radiation, δASR) and longwave [outgoing longwave radiation, $\delta(-\text{OLR})$] components are shown in the Right column. Positive fluxes indicate energy into the atmosphere. Blue and red histograms correspond to relative TC frequency as a function of latitude for the control and SWISHE experiments, respectively. The dashed red contour denotes the extent of 0.5 TC days per year. Shaded regions denote grid cells where all ensemble members feature the same sign of δR_{TOA} .

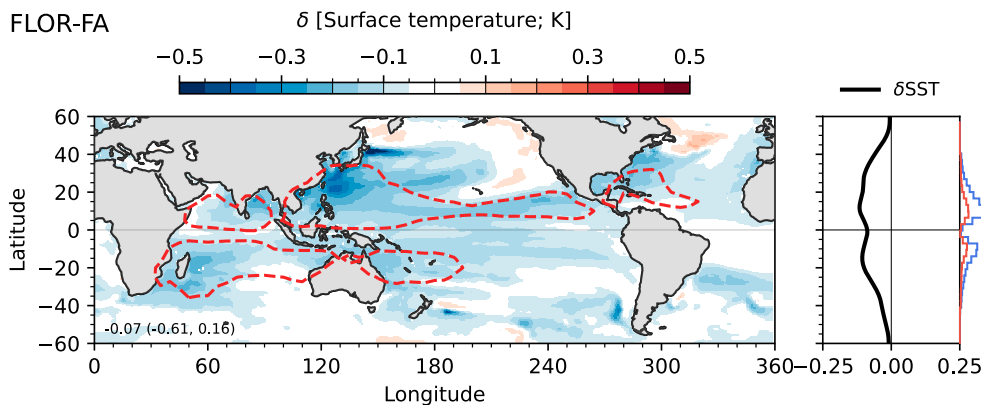


Fig. 3. As in Fig 2 for sea surface temperatures (δ SST).

suggests $\delta R_{\text{TOA, clear}} < 0$, as $\text{OLR}_{\text{clear}}$ is highly modulated by mid- and upper tropospheric moisture content (45, 47). The enhanced radiative cooling suggested by global mean δRH is reinforced by its similarity to changes in RH within climates in radiative-convective equilibrium undergoing convective aggregation (37, 48, 49).

Using RH as a proxy for atmospheric moisture, zonal mean $\delta \text{RH} < 0$ over TC-active latitudes (Fig. 4A, Left subplot), with maximal reductions in the Northern Hemisphere collocated with the maximal increase in zonal mean TC activity. $\delta \text{RH} < 0$ also occurs in the extratropical upper troposphere in both hemispheres, suggesting the influence of TC drying extends beyond cyclonically active basins.

The horizontal variability in δRH_{500} (Fig. 4B) reveals the role of TCs in drying the troposphere, as areas with a minimum of

0.5 TC days per year feature $\delta \text{RH}_{500} = -0.83 \pm 0.04\%$ relative to a lesser global $\delta \text{RH}_{500} < 0$ ($\delta \text{RH}_{500} = -0.13 \pm 0.01\%$). Unlike δRH_{500} in AGCM runs (SI Appendix, Fig. S4), free-tropospheric humidity changes are largely limited to TC-active basins, implying a spatial constraint on the extent of tropospheric drying to areas with $\delta \text{SST} < 0$. This constraint can be attributed to the regulation of SST-induced $\delta R_{\text{TOA}} > 0$ by radiative cooling from tropospheric drying in TC basins.

In comparison to mid and upper-tropospheric $\delta \text{RH} < 0$, global mean $\delta \text{RH} > 0$ in the boundary layer hints at increases in low cloud fraction, which may contribute to $\delta \text{SST} < 0$ by reducing downwelling solar radiation at the surface (10, 50, 51). However, increased low cloud fraction also leads to $\delta \text{OLR} > 0$ relative to a tropical climatology with higher cloud cover emitting at lower temperatures.

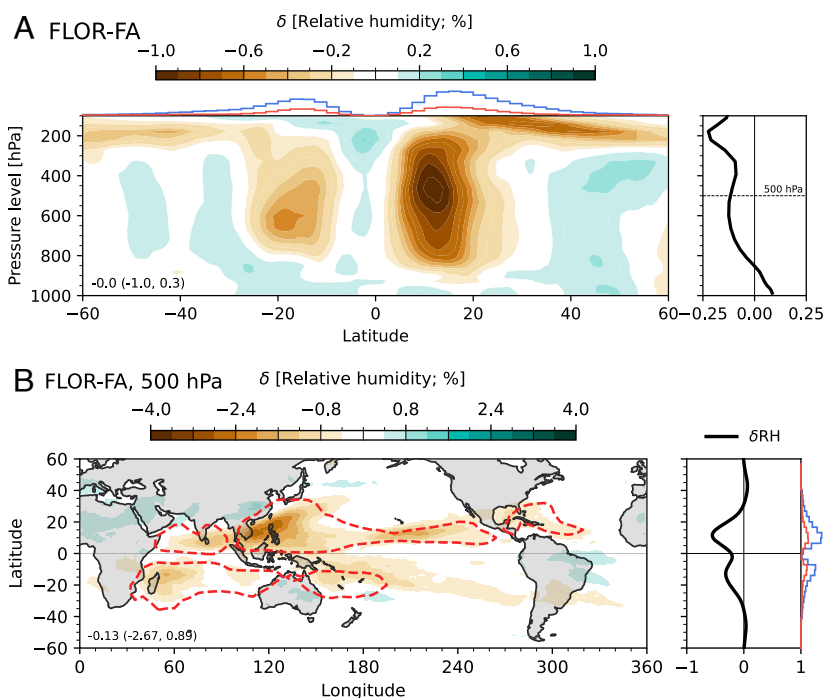


Fig. 4. Annual mean difference in relative humidity between control and SWISHE experiments for CM2.5-FLOR-FA model integrations. Zonally averaged time mean δRH is shown in the Left subplot of panel (A), with the corresponding global mean vertical profile of δRH in the Right subplot. Time-mean differences of mid-tropospheric relative humidity (i.e., δRH_{500}) are shown in the Left subplot of panel (B), with a corresponding zonally averaged time mean profile shown in the Right subplots of panel (B). Blue and red histograms correspond to relative TC frequency as a function of latitude for the control and SWISHE experiments, respectively. The dashed red contours in panel (B) denote the extent of 0.5 TC days per year. Shaded regions denote grid cells where all ensemble members feature the same sign of δRH .

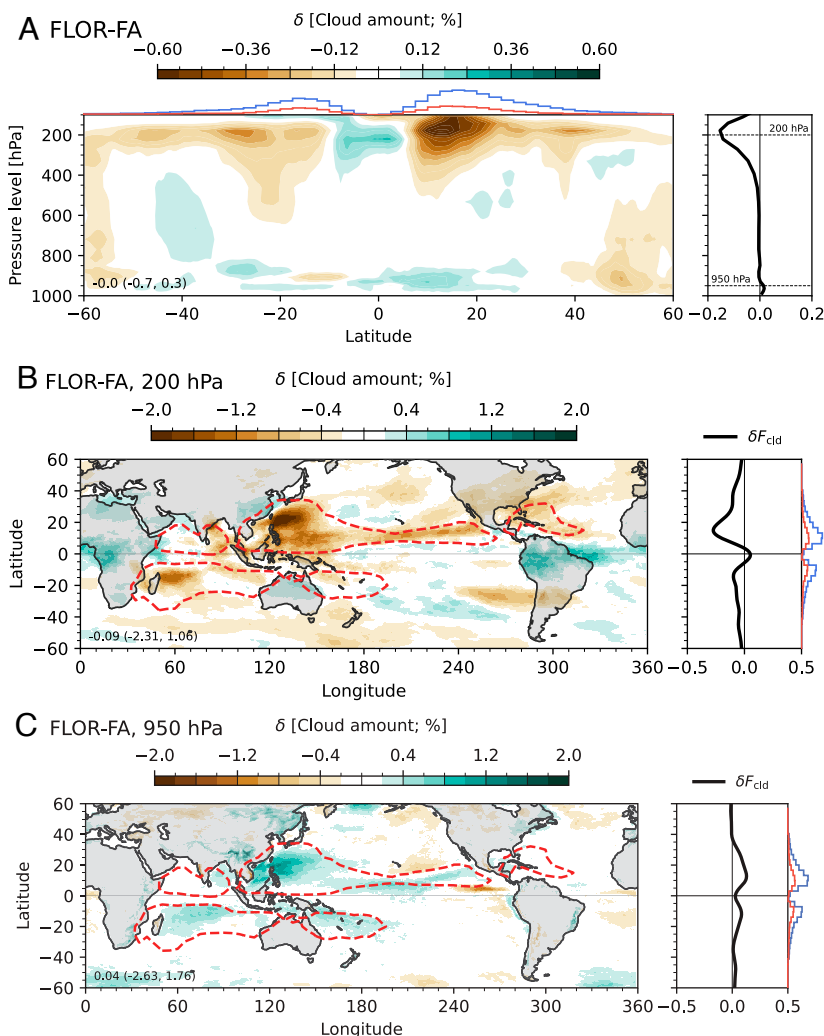


Fig. 5. Annual mean difference in cloud fraction between control and SWISHE experiments for CM2.5-FLOR-FA model integrations. Zonally averaged time mean δF_{cld} is shown in the *Left* subplot of panel (A), with the corresponding global mean vertical profile of δF_{cld} in the *Right* subplot. Time-mean differences of high cloud fraction (i.e., 200 hPa cloud fraction, $\delta F_{\text{cld}, 200}$) and low cloud fraction (i.e., 950 hPa cloud fraction, $\delta F_{\text{cld}, 950}$) are shown in the *Left* subplots of panels (B and C), respectively, with corresponding zonally averaged time mean profiles shown in the *Right* subplots of panels (B and C). Blue and red histograms correspond to relative TC frequency as a function of latitude for the control and SWISHE experiments, respectively. The dashed red contours in panels (B and C) denote the extent of 0.5 TC days per year. Shaded regions denote grid cells where all ensemble members feature the same sign of δF_{cld} .

Fewer High Clouds with More TCs. $\delta R_{\text{TOA, CRE}} < 0$ is explained by reductions in high cloud fraction with more TC activity in coupled GCM runs, despite modest increases in global mean low cloud fraction.

As shown in Fig. 5A, decreases in zonally averaged cloud fraction with greater TC activity (δF_{cld}) are maximal in the upper troposphere at latitudes with TC activity, particularly in the Northern Hemisphere where TCs most frequently occur. Reductions in zonal mean high cloud fraction also extend into the midlatitudes in both hemispheres. Using 200 hPa cloud fraction as a proxy for high cloud cover ($\delta F_{\text{cld}, 200}$, Fig. 5B), the distribution of $\delta F_{\text{cld}, 200}$ reveals a reduction of high cloud cover in areas commonly affected by recurring storms, indicating that TCs also decrease extratropical high cloud cover, albeit considerably less than in TC-active basins.

Similar in location to $\delta F_{\text{cld}, 200}$, increases in low cloud with greater TC activity (i.e., 950 hPa cloud fraction, $\delta F_{\text{cld}, 950}$) are maximal in cyclonically active latitudes (Fig. 5C). This aligns with the spatial distribution of δR_{TOA} and $\delta F_{\text{cld}, 200}$, further demonstrating the connections between radiative cooling and the basin-scale climatic influence of TC activity.

The mechanisms resulting in δF_{cld} differ between low and high clouds. $\delta F_{\text{cld}, 950} > 0$ where TCs occur, implying enhanced boundary layer moisture availability from TC-induced evaporation and low-level moisture convergence (52). Indeed, low-level relative humidity increases in areas with TC activity, indicating that increased low cloud fraction is at least partially a signature of increased TC activity. However, increased low cloud fraction outside of TC-active basins suggests a response to the SST adjustment and anomalous large-scale subsidence (10).

To examine the role of TC activity in the reduction of high clouds, upper-tropospheric divergence is investigated as a contributing mechanism. The TC secondary circulation (i.e., vertical-radial circulation) can be characterized by a concentrated region of deep convection surrounded by an extensive region of subsidence, resembling a convectively aggregated domain. Assuming mass conservation between TC and non-TC regions, anomalous ascent at TCs must be balanced by anomalous subsidence away from TCs. Given these assumptions, a constraint between mass and energy (3, 53) enables the relationship between changes in free-tropospheric vertical motion ($\omega_{\text{r, clear}}$, which can be related to divergence by continuity) and clear-sky radiative

cooling ($Q_{r, \text{clear}}$) to be investigated (see clear-sky thermodynamic relations in *SI Appendix*, Eq. S2). Focusing on TC-active regions, an increase in column vapor with more TCs implies $\delta Q_{r, \text{clear}} < 0$, such that $\delta \omega_{r, \text{clear}} < 0$ for a given static stability, S . Coupled GCM runs exhibit $\delta Q_{r, \text{clear}} < 0$ and $\delta \omega_{r, \text{clear}} < 0$ with maximum reductions occurring near 200 hPa (*SI Appendix*, Fig. S5 B and C), which is the level of maximum $\delta F_{\text{cld}} < 0$. As $\delta \omega_{r, \text{clear}} < 0$ occurs at a level where $\omega \gg 0$, the vertical gradient of $\omega_{r, \text{clear}}$ decreases, indicating a reduction in clear-sky radiative divergence from TC-active regions (*SI Appendix*, Fig. S5E). In other words, mass divergence out of TC-active regions decreases with enhanced TC activity. Therefore, with more TC activity, the decrease in convective outflow is associated with a reduction in high cloud fraction.

Summary and Discussion

Our findings highlight the ability of TC activity to induce large-scale radiative cooling in a coupled GCM by cooling SSTs, decreasing tropospheric humidity, and reducing high cloud cover. Notably, we find that large-scale radiative cooling occurs from the increase in longwave emissions exceeding the increase in absorbed solar radiation, which contrasts direct radiative cooling by TCs due to increased reflection of solar radiation exceeding decreased longwave emission.

TC Activity, Convective Aggregation, and Climate Sensitivity. A natural follow-up question to these results is: how do TCs affect climate sensitivity? This is a relevant question in the context of a warming climate, within which TC activity is expected to change (20, 54). Given the role of TC activity in organizing convection, the results herein show that a climate state with more TC activity features increased convective aggregation at the process-scale (as TCs are examples of aggregated convection) and at the basin-scale (due to the remote effects of TC activity). A growing body of work indicates that aggregation reduces climate sensitivity and may act as a stabilizing factor on the climate system (3, 48, 55, 56). The effect of TCs on climate sensitivity (λ_{TC}) may be estimated in the future by obtaining the TC radiative feedback: $\lambda_{\text{TC}} = (\delta R_{\text{TOA}}/\delta \text{TC}) (\delta \text{TC}/\delta T)$ (57). Assessing the linearity of δR_{TOA} to TC activity is possible by applying SWISHE to reduce TC activity continuously and to smaller, more realistic magnitudes, as compared to near-total suppression performed herein, although we leave this for future investigation. Although the findings in this study can be used to quantify $\delta R_{\text{TOA}}/\delta \text{TC}$, the response of TC activity to surface temperature changes ($\delta \text{TC}/\delta T$) is a critical quantity of great interest that is not yet known (20, 54, 58).

However, the significance of increasing aggregation on climate sensitivity remains uncertain (48). A key open question is the relative magnitude of any role of TCs on climate sensitivity relative to that of large-scale processes contributing to aggregation (i.e., changes to the Intertropical Convergence Zone through an iris effect) given the small size and transient nature of TC activity, or whether the extreme intensity of TCs leads to a consequential effect. Therefore, we shed light on the potential for climate-scale effects of TCs while contextualizing the scale of TC impacts within the spectrum of convective aggregation, whose effects on climate sensitivity require further research.

The potential for TC activity to act as a feedback on a warming climate may also be supported by TC-induced reductions in global mean surface temperature. Potential intensity theory predicts an increase in the theoretical upper limit of TC intensity with warmer SSTs (59). Although SST alone is an insufficient predictor of observed maximum TC intensity (60, 61), modeling

studies have shown strong correlations between SST and TC intensity (22, 62), suggesting that TCs will intensify assuming warmer SSTs. These findings expand on the literature analyzing localized TC impacts on SST (39–41, 63, 64) by evaluating TC impacts on larger spatial and longer time scales, showing that increases in TC frequency and intensity reduce global mean surface temperature and potentially regulate further increases to TC intensity by decreasing SSTs.

As mentioned previously, TC activity is expected to change with a warming climate. The findings in this study do not explicitly distinguish the fraction of δR_{TOA} due to changes in TC frequency or median TC intensity, although intense TCs (i.e., hurricane-strength storms) are suppressed to a higher degree than all TCs by SWISHE. We hypothesize that with more intense TCs, radiative cooling will intensify due to increasingly organized convection. Composite analysis of δR_{TOA} at TC using observational and GCM data suggests there is a negative relationship between TC intensity and δR_{TOA} , likely due to decreases in ASR with higher cloud tops (*SI Appendix*, Fig. S5), although further investigation is required to understand the relationship between remote δR_{TOA} and TC intensity. Consideration of the change in spatial extent of TC activity with global warming (i.e., a poleward shift) is also relevant, as radiative effects on the climate system are sensitive to location, particularly latitude.

TC Activity and Large-Scale Energy Transport. Although the influence of TC activity on climate has largely been discussed on a global scale, the upscale effects of TCs are shown to exhibit spatial variability, implying that TC activity modulates energy transport. We first focus on the implications of TC-induced large-scale energy transport by radiative cooling.

Radiative cooling induced by TCs exhibits both meridional and zonal asymmetry. Considering the meridional direction, zonal mean δR_{TOA} is minimal north of the equator (Fig. 2), implying energy convergence into the tropical Northern Hemisphere with more TCs. In conjunction with colocated δSST minima, it is implied that TC activity induces anomalous equatorward ocean heat transport, which aligns with studies showing increases to tropical ocean heat content due to enhanced upper ocean forcing similar in latitude and frequency to TCs (65–67). Additionally, zonal mean $\delta R_{\text{TOA}} < 0$ in the Northern Hemisphere, while $\delta R_{\text{TOA}} \sim 0$ in the Southern Hemisphere, indicating net cooling in the Northern Hemisphere, which suggests a southward ITCZ shift (68–70). Assuming that an equatorward shift in the ITCZ suppresses TC genesis (71–73), a self-imposed suppression of TC activity appears to emerge: Enhanced TC activity shifts the ITCZ equatorward, which suppresses TC activity, leading to a reduction in hemispheric asymmetry in radiative cooling. Given that TC frequency remains relatively constant with time in observations and model integrations used herein, the suppression must be balanced by mechanisms that enhance TC activity, although we leave these mechanisms as an area for future research.

Caveats and Future Work. These results feature several caveats that pave the way for future work. One caveat is the nominal grid spacing of GCMs (~ 50 km), which does not represent the full intensity distribution of TCs (detailed information about GCMs used in *Global Climate Models*). Consequently, the most energetic processes in the GCM runs are parameterized, which are expected to introduce differences between simulated and observed TC characteristics. Moreover, TC-related impacts on climatic processes in the GCMs explored here have been shown

to be sensitive to spatial resolution (74, 75). Although maximum TC intensities do not exceed 10 m wind speeds of $\sim 55 \text{ m s}^{-1}$ in the GCM runs analyzed herein (i.e., Category 3 on the Saffir–Simpson hurricane wind scale), the spatial distributions and frequency of TCs resemble the observed climatology (*SI Appendix, Fig. S1*). Because a goal of this study was to understand large-scale effects of a small-scale phenomenon, the chosen GCMs were selected to simulate multidecadal TC activity and global climate processes without prohibitive computational expense. However, given recent developments in modeling capabilities of the coupled atmosphere-ocean system at higher spatial resolution (76–78), future work can expand on the findings herein by investigating the role of TC activity over a larger intensity spectrum with fewer parameterizations for meso- and kilometer-scale energetic phenomena.

Additionally, longer model integrations ($> 300 \text{ y}$) may provide enhanced confidence in the role of TC activity on ocean processes at large-scale given the relative inertia of the ocean to perturbations relative to the atmosphere. Despite the lack of exploration of TC impacts on oceanic processes in this study, we note that δSST indicates TC-induced perturbations to ocean heat transport (26, 64, 79–81), which has significant implications for the climate state and merits further investigation. Another caveat is the usage of simulations to infer upscale influence of TC activity. Although improvements in observational techniques can improve the approximation of TC radiative effects (e.g., spatial resolution, increased availability of multispectral data, improved detection of TC clouds), quantifying the upscale influence of TC activity on the climate is not possible by observations, thus prompting the model-based approach used herein.

Materials and Methods

Global Climate Models. The GCMs used in this analysis are two atmosphere-only GCMs (AGCMs), AM2.5 and HiRAM, and a coupled atmosphere-ocean GCM, CM2.5-FLOR. CM2.5-FLOR-FA (Forecast Low-Ocean Resolution) (82) is a modification of the NOAA Geophysical Fluid Dynamics Laboratory (GFDL) coupled atmosphere-ocean GCM, CM2.5 (83), that modulates ocean resolution as a function of latitude to increase computational efficiency, while maintaining sufficient resolution to simulate sub-TC-scale perturbations ($\leq 1^\circ$) in the tropics and subtropics. The decreased ocean resolution relative to CM2.5 is adequate for the purpose of this study, which is focused on atmospheric phenomena (i.e., TCs). AM2.5 is the atmospheric component of GFDL coupled atmosphere-ocean GCM CM2.5. HiRAM (84) is modified from GFDL AM2.1, with significant modifications made (e.g., dynamical core change, increased vertical resolution, simplified cloud scheme). A key difference between HiRAM and AM2.5 is the cumulus parameterization, which features a single entraining and detraining plume as compared to the relaxed Arakawa-Schubert parameterization in CM2.5, which features a spectrum of entraining plumes. Both atmosphere-only GCMs are forced with fixed SSTs prescribed as a repeated monthly climatology computed from the United Kingdom's Met Office Hadley Centre Sea Ice and Sea Surface Temperature dataset (85) averaged from years 1986 to 2005.

Three CM2.5-FLOR-FA experiment ensemble members (i.e., configurations with the SWISHE algorithm) were implemented for comparison with the long-run control member using initial conditions from the control member at 50 y intervals from each other (44). Flux adjustment is implemented in CM2.5-FLOR-FA runs to reduce biases of mean SST and surface wind stress in CM2.5 relative to observations, resulting in improved representations of the mean climate state and TC activity relative to CM2.5 (82). Flux adjustment is imposed independently of surface fluxes modified by SWISHE and is unaffected by its implementation.

All GCMs feature spatial resolutions of $\sim 50 \text{ km}$ in the atmosphere, while CM2.5-FLOR-FA modulates ocean resolution from 1° at the poles to approximately 0.33° at the Equator. At this horizontal resolution, large-scale

convection (order $\sim 150 \text{ km}$) is explicitly resolved, where as small-scale convective processes are generally parameterized. Model integrations of 50 y and a minimum 100 y were obtained for the AGCMs and each CM2.5-FLOR-FA member, respectively. CM2.5-FLOR-FA was spun-up for 2000 y prior to the model integration period used for analysis to allow for climate system equilibration. TCs are tracked in all GCMs by identifying vortices using sea-level pressure anomalies, horizontal anomalies of 200 to 350 hPa temperature to identify a warm core, 10 m horizontal wind speeds, and 850 hPa relative vorticity exceeding a magnitude of $1.5 \times 10^{-4} \text{ s}^{-1}$ consecutively for a minimum of 72 h (86, 87).

Modulation of TC Activity in Climate Models. SWISHE (Suppressed Wind-Induced Surface Heat Exchange) is an algorithm implemented in the GCMs used for this study to reduce TC activity (44). SWISHE is conditionally implemented such that latent heat fluxes (LHF) are suppressed at grid cells experiencing TC-like conditions. The SWISHE algorithm is designed to target grid cells with TC-like conditions while avoiding the suppression of LHF at grid cells experiencing similar, albeit weaker, atmospheric phenomena (e.g., South Asian monsoon, strong mesoscale convective systems). Although a graded variation in TC suppression to smaller magnitudes would reflect more realistic variability in TC frequency than complete suppression of SWISHE, we employ complete suppression as a hypothetical to induce a strong climate response, such that the response to the perturbation is attributable to TC activity.

Implementation is determined by field values at each grid cell, both at the surface and throughout the atmospheric column corresponding to the given grid cell. Field value conditions for implementation are chosen to represent properties characteristic of the inner cores of TCs (e.g., 10 m wind speeds exceeding 15 m s^{-1} , 850 hPa relative vorticity magnitudes exceeding $1 \times 10^{-4} \text{ s}^{-1}$, lower and mid-tropospheric relative humidity values exceeding 60%). Additionally, SWISHE is implemented only on seawater grid cells and those experiencing SSTs exceeding 293 K. SWISHE is implemented on each GCM physics timestep such that LHF are suppressed during model integration.

Using SWISHE in GCMs, perturbed climate states are generated with multimodel mean reductions in TC frequency of 59% for all TCs in the perturbed climate states relative to their corresponding controls, with a 86% decrease in the number of hurricane-strength TCs (i.e., maximum 10 m wind speeds exceed 32.5 m s^{-1}). Statistics for individual models' runs are available in *SI Appendix, Table S1*.

Suppression of TC activity occurs roughly proportionally to TC activity in each basin (*Fig. 1C* and *SI Appendix, Fig. S1 G and J*), with the strongest magnitude of suppression occurring in the western north Pacific basin ($\geq 4 \text{ d per year}$). All GCMs reproduce TC frequencies and distributions similar to observations, with peak TC activity in the western North Pacific Ocean and substantial activity in every TC-active basin. More details about GCM ability to reproduce the observed distribution and frequency of TCs can be found in *SI Appendix, Fig. S1*. The impacts of evaporation directly suppressed by SWISHE relative to total change in evaporation between control and SWISHE configurations are limited and discussed in detail in *SI Appendix*.

Data, Materials, and Software Availability. Dataset has been deposited in Zenodo (DOI: [10.5281/zenodo.17154138](https://doi.org/10.5281/zenodo.17154138)) (87).

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